E350

U1300 and U3400

RELEASE NOTES FOR F/W VERSION 7.0

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Revision history

Version	Date	Comments
1	24/01/2020	Initial Release

Important Information
This document provides information relating to E350 U1300 Series 1, Series 2 and U3400 meter firmware (the "Product")

Landis & Gyr Pty Ltd (ABN 78 002 894 224) trading as Landis+Gyr.

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Table of contents

1 About this document 4
1.1 Purpose & Scope 4
1.2 Definitions, Acronyms, and Abbreviations 4
1.3 References 4
2 About the release, compatible products and Summary 5
2.1 New Features for Supported Meters 6
2.1.1 General New Features 6
2.1.2 Firmware Upgrade 6
2.1.3 Firmware Downgrade 6
2.1.4 QOS Current monitoring (Sag and Swell) 6
2.1.5 Load Control random delay methods 7
2.1.6 Four Load Control Seasons 8
2.1.7 New Display Items. 8
2.1.7.1 Temperature Display Item. 8
2.1.7.2 Last SSN Antenna Switch Request Display Item. 8
2.1.7.3 E35C Module Status Display Item. 9
2.1.7.4 E35C Module Signal Strength Display Item. 9
2.2 Load Profile Intervals. 10
2.2.1 1 Minute Load Profile Intervals. 10
2.2.2 Temperature Load Profile Items. 10
2.2.3 E35C Module Signal Strength Load Profile Item. 10
2.2.4 E35C Module Status Load Profile Item. 11
2.3 Itron MV90, MVRS, IEE Support 11
2.4 Command Center 12
2.4.1 Command Center Software Quality Assurance testing 12
2.5 Bug fixes in this release 12
2.6 Known Issues in this release 12

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1 About this document

1.1 Purpose & Scope

The purpose of this ***6.0 Firmware Release Note*** is to communicate new features, limitations and changes in meter functionality.

The 7.0 firmware release is for the E350 - U1300 and U3400 meters.

1.2 Definitions, Acronyms, and Abbreviations

RSM:	Relay State Machine for the load control relay
LC:	Load Control
LP:	Load Profile
CLC/R:	Controlled Load Contactor or Relay
CSM:	Customer Supply Monitoring including Neutral Integrity
QoS:	Quality of Supply
AMI:	Advanced Metering Infrastructure
AD:	Auto Disconnect
S2:	Series 2
SCC:	Supply Capacity Control
STS:	Smart Time Synchronisation
HES:	Head End Systems
E350:	Family name of Landis+Gyr modular meter including the types U1200, U1300, U1300 S2 U3300, U3400 and U3350
E650:	Family name of Landis+Gyr commercial and industrial meter including the type EM5400 and EM5400 S2
E420:	Family name of Landis+Gyr integrated meter S1100 RF and S1100 LTE
R1100:	Landis+Gyr’s smart load control device.

1.3 References

E350 U1300 User Manual

E350 U3400 User Manual

EMPWin+ User Manual

2 About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released firmware version: 6.0

This Released Version: 7.0

Build Version: 006 (U3400) and 006 (U1300)

Compatible Meter	Firmware Upgrade*	Firmware Downgrade*
U1300	4.2 → 5.4 (Series 1 Only) ↓ 4.3 → 5.4 (Series 1 Only) ↓ 4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.8 → 6.0 ↓ 4.9 → 6.0 ↓ 5.0 → 6.0 ↓ 5.4 → 6.0 ↓ 6.0 → 7.0	7.0 → 6.0
U3400	4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.7 → 5.4 ↓ 4.9 → 6.0 ↓ 5.0 → 6.0 ↓ 5.4 → 6.0 ↓ 6.0 → 7.0	7.0 → 6.0

Important Note: *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please also refer to the intermediary firmware release notes when skipping meter firmware version.

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1, Series 2, 16Mb and 64Mb Variants)
- E350 U3400 meters (16Mb and 64Mb Variants)
- EMPWin+ 3.7.27

The FT100 release 5.4 or earlier must not be used with 7.0 meter firmware.

EMPWin+ 3.7.27 or greater must be used with 7.0 meter firmware.

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2.1 New Features for Supported Meters

2.1.1 General New Features

The E350 U1300 & U3400 meters with 7.0 firmware now have the following enhancements.

- QOS current recording
- New Load Control randomisation methods.
- Four Load Control Seasons in the meter and ripple program.
- Last SSN Antenna Switch Request Display Item
- Signal Strength recording and display for the L35C 3G, LTE & RF
- 1 minute interval period
- Temperature recording and display

2.1.2 Firmware Upgrade

- The E350 U1300 & U3400 meters with 16Mb memory with firmware R5.4 or earlier are required to be upgraded to 6.0 and reprogrammed before upgrading the meter to 7.0.
- Meters running 6.0 can be upgraded to 7.0 F/W without full programming the meter, however **the passwords should be reset to allow access to the new Ripple season LC parameters using MeterComm 3.7.27 or later.**
- If new features in 7.0 F/W are required, then the meter must also be reprogrammed.
- U1300 and U3400 meters with 64Mb memory will be manufactured with 6.0 or 7.0 firmware.

2.1.3 Firmware Downgrade

Downgrade back from 7.0 to 6.0 is supported but any features that are unique to 7.0 will become displayed after the downgrade. Downgrade to any earlier version than 6.0 is not supported. Doing so might require the meter being to be returned to Landis & Gyr to be factory reprogrammed with the F/W version and program that is required. There is no automatic protection in the firmware that would prevent a customer from performing this type of data destructive operation.

2.1.4 QOS Current monitoring (Sag and Swell)

The U1300 now supports 3 QOS channels and the U3400 supports 6 QOS channels When selecting a negative value, this represents export current and when selecting positive values this represents import current. When selecting two negative values, the larger value is entered into the “Low Threshold“ and the smaller value is entered into the “High Threshold“.

The default Low Threshold for current is -100 and maximum is 100, these values basically disable current monitoring unless the export or import current exceed 100 amps.

Quality Of Supply

Number of Channels 6

QOS Channels

	QOS Quantity	Breach Time (seconds)	Low Threshhold	High Threshhold	Max Events Recorded per day
1 .	Current Phase A	20	-45.00	80.00	10
2 .	Current Phase B	20	1.00	80.00	10
3 .	Current Phase C	20	-45.00	-20.00	10
4 .	Voltage Phase A	20	160	300	10
5 .	Voltage Phase B	20	160	300	10
6 .	Voltage Phase C	20	160	300	10

The extra QOS items have been designed to allow a single entry to monitor both import and export current based on their own threshold levels. This has been made possible by using negative threshold current values to represent exporting current and for positive values to represent importing current.
The F/W will check if the current is importing or exporting on the phase selected and will use an absolute value comparison to determine if it is outside of the threshold limit. It is possible for a single QOS entry to check both import and export current on the selected phase. This would be achieved by setting the negative (export value) in the Low Threshold section and the positive (importing value) in the High Threshold area. For example, Low = -45 and high = 80 would result in a QOS event occurring if the current was exporting more than 45 amps or if the importing current exceeded 80 amps.
The QOS Statistics area has been modified to show up to three QOS sets of results and counters.

2.1.5 Load Control random delay methods

The U1300 & U3400 now supports three selectable random delay methods.

- On Power Up (traditional method - sets the random value on every power up)
- On Full Programming
- On Every Random Switch operation.

LC random delays are used in the Priority Override command, Under Frequency delay after event ends, Time Scheduler LC delays and Cyclic LC operation delays.

The three methods of random delay are also available in the ripple program.

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2.1.6 Four Load Control Seasons

The U1300 & U3400 now supports four Load Control Seasons that can be configured in the main meter program and in the ripple program. When selecting Extended 1 or 2 memory sizes with meter F/W 07.00.006 or later it is now possible to define four Seasons with different monitoring options. The dates for the four seasons are defined on the Controlled Load – Time Table Settings page. If Standard memory size is selected, then only two seasons are offered.

2.1.7 New Display Items.

The display list for the U1300 & U3400 now includes the Last SSN Antenna Switch Request, E35C Module Status, most recent Signal Strength and Temperature. The Signal Strength and Module Status features are available when using communication module firmware version 16.50 or greater.

2.1.7.1 Temperature Display Item.

The display list for the U1300 and U3400 now includes Temperature (per phase / element). The U1300 and U3400 have the following sensor options for the Display.

Temperature Sensor	U1300	U3400
Temperature Sensor A	Yes	Yes
Temperature Sensor B	Yes	Yes
Temperature Sensor C	N/A	Yes

2.1.7.2 Last SSN Antenna Switch Request Display Item.

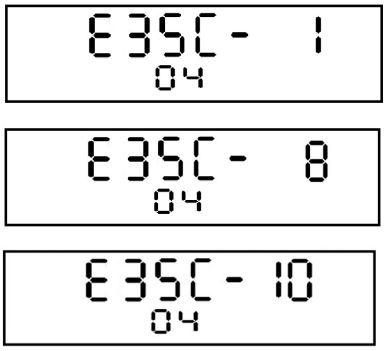
The display list for the U1300 and U3400 now includes the Last SSN Antenna Switch Request performed by the FT100.

2.1.7.3 E35C Module Status Display Item.

The display list for the U1300 and U3400 now includes the E35C Module Status. This status provides the user with an index number on the LCD that represents the Antenna switch position, Module Type installed, the mode its operating in and the signal strength unit.

Index	Antenna	Module Type	Mode	Unit
1	INT	E35C-RF	RF	RSSI
2	EXT	E35C-RF	RF	RSSI
3	INT	E35C-3G	3G	RSCP
4	EXT	E35C-3G	3G	RSCP
5	INT	E35C-LTE1	4G	RSRP
6	EXT	E35C-LTE1	4G	RSRP
7	INT	E35C-LTE1	3G	RSCP
8	EXT	E35C-LTE1	3G	RSCP
9	INT	E35C-LTEM	CAT-M	RSRP
10	EXT	E35C-LTEM	CAT-M	RSRP

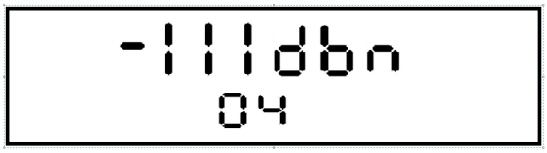
U1300 and U3400 Module Status display examples.



In case the communication module is not attached to meter and was unable to update the meter with the module status, the meter display will show: "E35C-Un".

2.1.7.4 E35C Module Signal Strength Display Item.

The U1300 and U3400 can display the most recent Signal Strength.
The signal strength is measured in decibels (dBm) and the closer the value is to 0, the stronger the signal.



Module Type	Signal Strength Range	Recommended Minimum signal level
E35C RF	-127 dBm to 0 dBm	- 95 dBm RSSI
E35C 3G	-113 dBm to -51 dBm	- 90 dBm RSCP
E35C LTE	-139 dBm to -44dBm	-103 dBm RSRP

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2.1.8 Load Profile Intervals.

The U1300 & U3400 meters now supports:

- 1 minute load profile intervals.
- E35C Module Status
- Most recent Signal Strength
- Temperature.

The Signal Strength and Module Status features are available when using communication module firmware version 16.50 or greater.

2.1.8.1 1 Minute Load Profile Intervals.

The U1300 & U3400 now supports 1 minute load profile intervals for meters with Extended 1 or Extended 2 Memory size. If a single LP group is selected with 1-minute intervals, then up to four channels can support for 44 days for Extended 1 memory option. For Extended 2 memory meters 439 days of 1-minute LP is supported. Standard memory size option only supports 10,15, 30 & 60 minute intervals.

2.1.8.2 Temperature Load Profile Items.

The U1300 and U3400 meters can now load profile Temperature (per phase / element). Load Profile can be recorded as Instantaneous (at end of interval) or Minimum, Maximum or Average for the LP interval. Temperature values are updated once per minute.

The following sensor options are available.

Temperature Sensor	U1300	U3400
Temperature Sensor A	Yes	Yes
Temperature Sensor B	Yes	Yes
Temperature Sensor C	N/A	Yes

2.1.8.3 E35C Module Signal Strength Load Profile Item.

The U1300 and U3400 can now load profile the most recent Signal Strength.
The signal strength is measured in decibels (dBm) and can range from approximately -44 dBm to -139 dBm. The closer that number is to 0, the stronger the signal. The signal strength is updated in the meter every minute by the E35C communication module, when using communication module firmware version 16.50 or greater.
Cellular signal values are sampled every 15 seconds and the meter averages these samples over a 3 minute period, this produces a smoothing of the reported values. LP can be recorded as Instantaneous (at end of interval) or Minimum, Maximum or Average for the LP interval period. The RF signal strength values are extracted from the packet data arriving at the meter and there is no averaging or smoothing of values so a wider range of values could be reported in the LP data.
If comms is lost or unavailable, then the meter will continue to use the last reported value in LP values and for the displayed value.

2.1.8.4 E35C Module Status Load Profile Item.

The U1300 and U3400 can now load profile the E35C Module Status. This status provides the user with the status of the Antenna switch position, Module Type installed, the mode its operating in and the signal strength unit.

C:\Users\sethikhy\Desktop\EMPWin+ 900.1.749 - Internal\U1300SAM180456_3.mlr [U1300] - Meter Reading Viewer

File View Help

Datetime	kWh Imported Total	kWh Exported Total	Maximum Signal Strength	Module Status
18/11/2019 17:46	0.000	0.017	-113 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:47	0.000	0.017	-104 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:48	0.000	0.017	-116 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:49	0.000	0.017	-109 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:50	0.000	0.017	-116 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:51	0.000	0.017	-118 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:52	0.000	0.017	-113 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:53	0.000	0.017	-101 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:54	0.000	0.017	-100 dBm	INT-E35C-RF-RF-RSSI
18/11/2019 17:55	0.000	0.017	-107 dBm	INT-E35C-RF-RF-RSSI

Module Status: INT-E35C-RF-RF-RSSI

Antenna Switch Position
- Internal (INT)
- External (EXT)

Module Type
- E35C-RF
- E35C-3G
- E35C-LTE1
- E35C-LTEM

Operating Mode
- RF
- 3G
- 4G
- CAT-M

Unit of Measurement
- RSSI
- RSCP
- RSRP

2.2 Itron MV90, MVRS, IEE Support

MV90 has been tested with 7.0 F/W

- MV90 reads the whole 200 self-read records but will only report the most recent 8.
- MV90 can support the reading of 5 minute LP data.
- HHF files created by MeterViewer can be imported into MV90 with 5 minute LP data.
- MeterViewer HHF format can export up to the first 16 LP channels in a single LP group.

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2.3 Command Center

2.3.1 Command Center Software Quality Assurance testing.

Command Center (CC) version 7.4.1.15 has been tested with 7.0 F/W.

The following functionality is not supported by CC 7.4.1.15 and will be available in CC 7.6:

- 1 minute load profile intervals.
- E35C Module Status
- Most recent Signal Strength
- Temperature.

2.4 Bug fixes in this release

Item No.	*Description*
TFS-856985 **TFS-857025** **TFS-43102** **TFS-1567** **Mantis-3904** **TFS-1067** **TFS-43106**	With firmware versions 6.0 and earlier, If the Ripple Power Restore option was set to Time Table AND the random power restore range has been set to 0 then the Time Table schedule will not run. It is possible that after installation the meter might not switch the LC relay when the first Ripple Telegram is received. If the meter has been programmed with Ripple, and a power outage occurs before the first valid telegram is sent, then the LC relay might not switch. Subsequent ripple telegrams are handled correctly. This issue is unlikely to occur because the random power restore is not likely to be set to 0 by customers. However, Meter firmware 7.0 has been enhanced to resolve this issue.

2.5 Known Issues in this release

Item No.	*Description*
FTS-857014	Missing Event log 2118 (Extended Load Control) for Four Season LC Change in Ripple Time Table. This is only a missing event and will be fixed in a future F/W release.
Mantis-3584	If there is a power outage across a LC season change from LC season 1 to LC season 0 then the event that indicates this LC season change to 0 is not recorded. The logging works correctly for LC season change from 0 to 1 across a power outage. The LC scheduler works correctly it is only the event that is missing.
TFS-48967 **TFS-857816** **Mantis-3906**	<u>Demand Limit Load Control</u> If Under Frequency Load Control (UFLC) module is triggered due to bad frequency, then the frequency recovers and UFLC starts its Nominal Delay period AND then a short power cycle occurs (about 3 seconds) AND on Power Up the Frequency is still within Normal limits; under these conditions the UFLC module might not correctly exit its Nominal waiting period. This can cause the UFLC module to block other lower priority RSM commands such as Ripple telegrams. It appears that when this incorrect state is reached that it might require two power cycles to correctly reset the RSM logic.

TFS-857051 **TFS-48970** **TFS-1565** **Mantis-3918**	It has been observed that U1300 meters, the LC1 & LC2 random delay closed values tend to have the same minutes and seconds with only the hours being different. Random delay is not random enough for the two relays.
TFS-857835 **TFS-1566** **TFS-48701** **Mantis-3919**	It was noticed the Ripple Control (RC) icon on the LCD may not display correctly under the following scenario. The RC icon on the display should be slowly flashing if the meter contains a valid ripple program and it is in Failsafe mode. Writing meter program with Ripple support "False" settings to meter while it is in Failsafe mode should automatically cancel any ripple operation and set RC icon to permanent On state. It was noticed the RC icon was not set to permanent On state and continued flashing slowly.

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